

Xuwan Yin

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[LinkedIn](#) · [GitHub](#) · [Google Scholar](#)

Education

2016 – 2022	Ph.D., Computer Engineering, University of Virginia, United States Ph.D. thesis: Generative Modeling With Adversarial Training Supervisor: Prof. Gustavo Rohde Research focus: adversarial machine learning, energy-based generative models, multimodal models, optimal transport in signal processing
2011 – 2014	M.Eng., Computer Engineering, University of Science and Technology Beijing, China Master thesis: Robust Text Detection in Natural Scene Images Supervisor: Prof. Xu-Cheng Yin
2007 – 2011	B.Eng., Computer Science and Engineering, University of Science and Technology Beijing, China

Employment

2025 – Present	Independent Researcher, San Francisco / Qingdao Scalable energy-based generative models. Developed an adversarial training framework that unifies generation and classification in a single model, the first to scale energy-based models to ImageNet 256×256 (ICLR 2026).
2023 – 2025	Research Engineer, Center for AI Safety, San Francisco, United States Research on LLM safety evaluation, automated red teaming, adversarial robustness, and emergent value systems in AI. Contributed to HarmBench (ICML 2024), Utility Engineering (NeurIPS 2025), MASK Benchmark (NeurIPS 2026), Representation Engineering , and Unforeseen Adversaries .
2014 – 2015	Computer Vision Engineer, Yuanli Education Technology, Beijing, China Text detection and recognition in natural images.

Research Interests

Energy-based generative models: scalable training via adversarial training, text-to-image generation, and hybrid discriminative-generative models.

Machine learning robustness and security: adversarially robust image models and LLMs, safety evaluation and red-teaming of LLMs, and adversarial example detection.

Publications [\[Google Scholar\]](#)

Conference Papers

- The MASK Benchmark: Disentangling Honesty From Accuracy in AI Systems
Richard Ren, Arunim Agarwal, . . . , **Xuwan Yin**, . . . , Dan Hendrycks
NeurIPS Evaluations and Datasets Track (NeurIPS 2026) [[arXiv](#)]
- Scalable Energy-Based Models via Adversarial Training: Unifying Discrimination and Generation
Xuwan Yin, Claire Zhang, Julie Steele, Nir Shavit, Tony T. Wang
International Conference on Learning Representations (ICLR 2026) [[arXiv](#)]

- Utility Engineering: Analyzing and Controlling Emergent Value Systems in AIs
Mantas Mazeika, **Xu Wang Yin**, . . . , Dan Hendrycks
Conference on Neural Information Processing Systems (NeurIPS 2025) [[arXiv](#)] [[code](#)]
- HarmBench: A Standardized Evaluation Framework for Automated Red Teaming and Robust Refusal
Mantas Mazeika, Long Phan, **Xu Wang Yin**, . . . , Dan Hendrycks
International Conference on Machine Learning (ICML 2024) [[arXiv](#)] [[project](#)]
- Learning Energy-Based Models With Adversarial Training
Xu Wang Yin, Shiyong Li, Gustavo K. Rohde
European Conference on Computer Vision (ECCV 2022) [[arXiv](#)] [[code](#)] [[poster](#)] [[video](#)]
- GAT: Generative Adversarial Training for Adversarial Example Detection and Robust Classification
Xu Wang Yin, Soheil Kolouri, Gustavo K. Rohde
International Conference on Learning Representations (ICLR 2020) [[arXiv](#)] [[code](#)]
- OBJ2TEXT: Generating Visually Descriptive Language from Object Layouts
Xu Wang Yin and Vicente Ordonez
Empirical Methods in Natural Language Processing (EMNLP 2017, oral) [[paper](#)] [[code](#)]
- Effective Text Localization in Natural Scene Images with MSER, Geometry-Based Grouping and AdaBoost
Xu Wang Yin, Xu-Cheng Yin, Hong-Wei Hao, Khalid Iqbal
International Conference on Pattern Recognition (ICPR 2012) [[paper](#)]

Journal Articles

- End-to-End Signal Classification in Signed Cumulative Distribution Transform Space
AMH Rubaiyat, Shiyong Li, **Xu Wang Yin**, Mohammad Shifat E Rabbi, Yan Zhuang, Gustavo K. Rohde
IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI 2023) [[arXiv](#)]
- A Digital Camera-based Eye Movement Assessment Method for NeuroEye Examination
Mohamed Abul Hassan, **Xu Wang Yin***, Yan Zhuang*, Chad M Aldridge, Timothy McMurry, Andrew M Southerland, Gustavo K Rohde (*equal contribution)
IEEE Journal of Biomedical and Health Informatics (JBHI 2023) [[paper](#)]
- Radon Cumulative Distribution Transform Subspace Modeling for Image Classification
M Shifat-E-Rabbi, **Xu Wang Yin**, Abu Hasnat Mohammad Rubaiyat, Shiyong Li, Soheil Kolouri, Akram Aldroubi, Jonathan M. Nichols, Gustavo K. Rohde
Journal of Mathematical Imaging and Vision (JMIV 2021) [[paper](#)] [[code](#)]
- Neural Networks, Hypersurfaces, and Radon Transforms
Soheil Kolouri, **Xu Wang Yin**, Gustavo K. Rohde
IEEE Signal Processing Magazine (SPM 2020) [[paper](#)] [[arXiv](#)] [[code](#)]
- Robust Text Detection in Natural Scene Images
Xu-Cheng Yin, **Xu Wang Yin**, Kaizhu Huang, Hong-Wei Hao
IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI 2014) [[paper](#)]

Preprints & Workshop Papers

- RenderAttack: Hundreds of Adversarial Attacks Through Differentiable Texture Generation
Dron Hazra, Alex Bie, Mantas Mazeika, **Xu Wang Yin**, Andy Zou, Dan Hendrycks, Maximilian Kaufmann
NeurIPS Workshop on New Frontiers in Adversarial Machine Learning (NeurIPS 2024 Workshop)
- Representation Engineering: A Top-Down Approach to AI Transparency
Andy Zou, Long Phan, . . . , **Xu Wang Yin**, . . . , Dan Hendrycks
arXiv preprint (2023) [[arXiv](#)]

- Testing Robustness Against Unforeseen Adversaries
Max Kaufmann, Daniel Kang, . . . , **Xuwang Yin**, . . . , Dan Hendrycks
arXiv preprint (2023) [[arXiv](#)] [[code](#)]

Awards

2022		Top Reviewer , NeurIPS 2022
2014		Outstanding Graduate of Beijing, Beijing Municipal Commission of Education
2013		Winner , ICDAR 2013 Robust Reading Competition (15 teams from 7 countries)

Professional Service

Reviewer for NeurIPS 2022, 2024, 2025; ICLR 2022, 2026; ECCV 2022; ACL 2021, 2022, 2024; EMNLP 2021; and the journal *Neurocomputing*.